Power input (USB-C) and regulators

File: lcd_dual_power.kicad_sch

HDMI receiver LT8619C

File: lcd_dual_hdmi.kicad_sch

AT070TN92 panel, bias supplies, VCOM, backlight

File: lcd_dual_panel.kicad_sch

STM32F072 controller and GT915 touch

File: lcd_dual_mcu.kicad_sch

AT070TN92 7" 800x480 TTL panel driver: HDMI (LT8619C) + GT915 touch -> USB HID (STM32F072).

Power: USB-C 5 V (from Raspberry Pi USB port, ~0.7 A max with backlight at 100 %).

U10: STM32F072CBT6 or CH32X035C8T6 (same LQFP-48 footprint). SW2 = CH32 USB bootloader (D+ high at power-up).

Panel power-up order (MCU controlled): DVDD -> RESET -> VGL/AVDD -> VGH -> data -> backlight.

J3 pad numbers equal AT070TN92 FPC pin numbers; use a TOP-contact 0.5 mm 50-pin FPC connector.

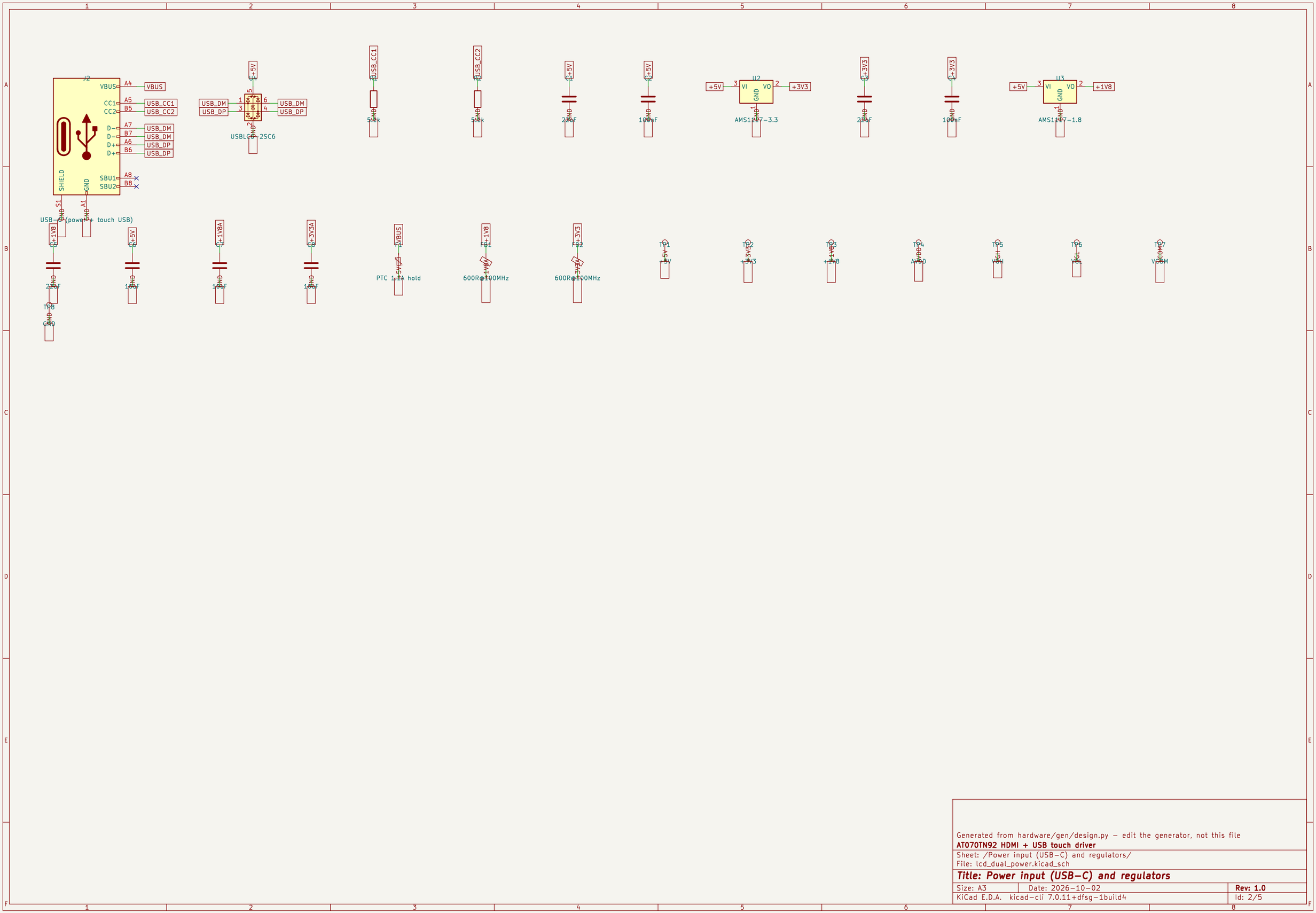
VCOM (pins 6/46) is adjustable with RV1 (2.55..4.5 V): trim for minimum flicker.

LT8619C HDMI->RGB888, TPS61040 bias, TPS61169 backlight, STM32F072 / CH32X035 USB touch

Sheet: /
File: lcd_dual.kicad_sch

**Title: AT070TN92 HDMI + USB-touch driver board**

| Size: A4 | Date: 2026-10-02 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 1/5 |



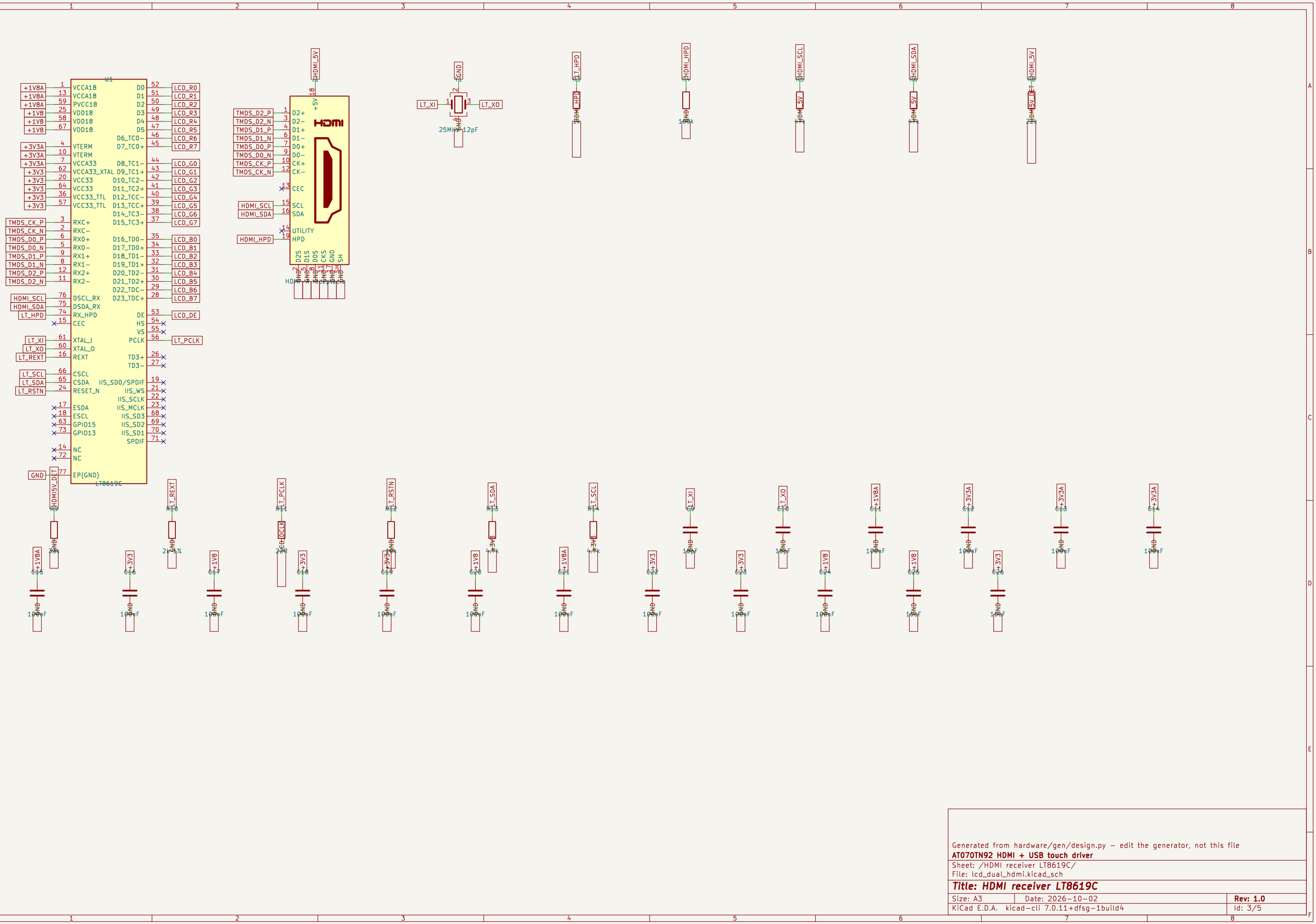

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**AT070TN92 HDMI + USB touch driver**

Sheet: /Power input (USB-C) and regulators/
File: lcd_dual_power.kicad_sch

***Title: Power input (USB-C) and regulators***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-02 | **Rev: 1.0** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 2/5 |




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**AT070TN92 HDMI + USB touch driver**

Sheet: /HDMI receiver LT8619C/
File: lcd_dual_hdmi.kicad_sch

**Title: HDMI receiver LT8619C**

| Size: A3 | Date: 2026-10-02 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 3/5 |

AT070TN92 FPC (0.5mm 50P, top contact)

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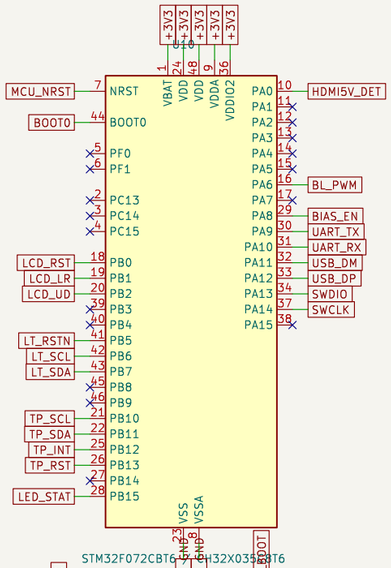
**AT070TN92 HDMI + USB touch driver**

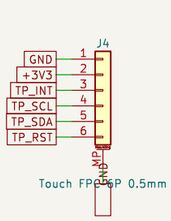
Sheet: /AT070TN92 panel, bias supplies, VCOM, backlight/

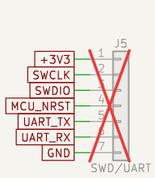
File: lcd_dual_panel.kicad_sch

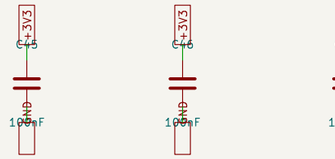
***Title: AT070TN92 panel, bias supplies, VCOM, backlight***

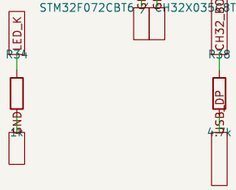
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|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 4/5 |

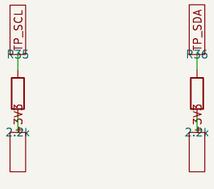















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**AT070TN92 HDMI + USB touch driver**

Sheet: /STM32F072 controller and GT915 touch/
File: lcd_dual_mcu.kicad_sch

***Title: STM32F072 controller and GT915 touch***

| Size: A3 | Date: 2026-10-02 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 5/5 |